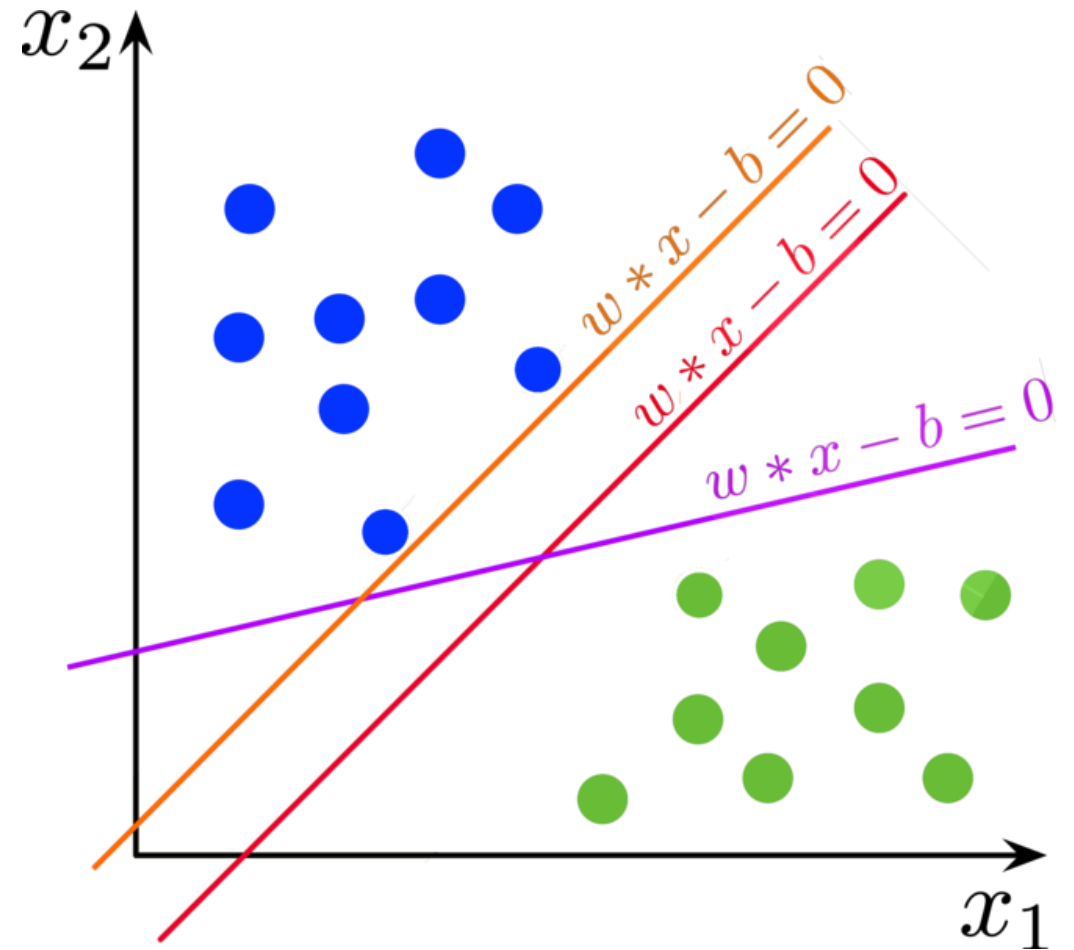


Support Vector Machines

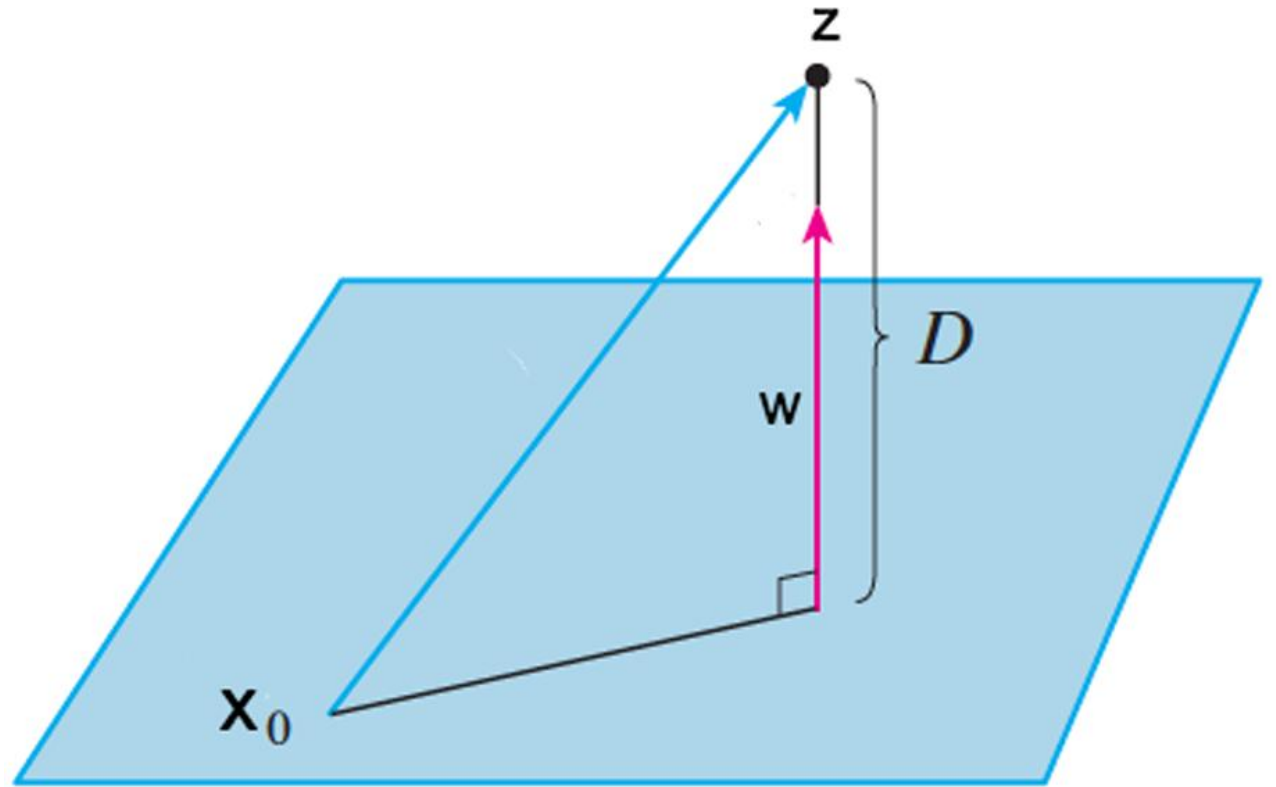
Perceptron

$$\forall i, \quad \begin{cases} w^T x_i - b \geq 0, & y_i = 1 \\ w^T x_i - b \leq 0, & y_i = -1 \end{cases}$$



Distance to Hyperplane

$$D = \frac{|w^T z - b|}{\|w\|_2}$$



Hard-Margin SVMs

$$\max_{w,b} m, \quad m \geq 0$$

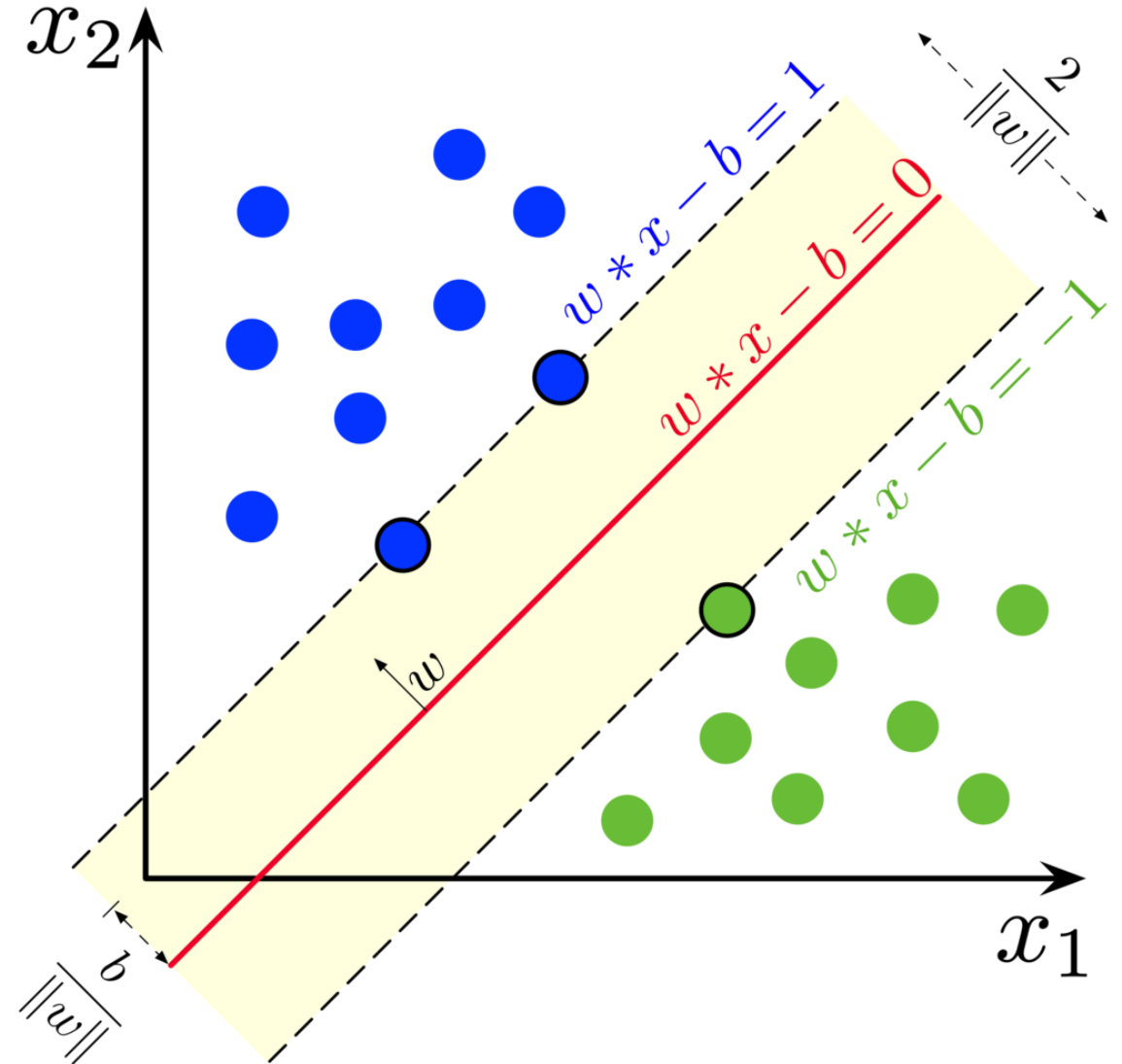
$$y_i \frac{(w^T x_i - b)}{\|w\|_2} \geq m \quad \forall i$$

But if (m^*, w^*, b^*) is a solution, then

$(m^*, \alpha w^*, \alpha b^*)$ is also solution $\forall \alpha > 0$

$$\max_{w,b} \frac{1}{\|w\|_2}$$

$$y_i (w^T x_i - b) \geq 1 \quad \forall i$$



Optimization Task

$$\begin{cases} \frac{1}{2} w^T w \rightarrow \min \\ y_i (w^T x_i - b) \geq 1 \end{cases}$$

Primal problem solution with CVXOPT Package

Task that QP solver solves:

$$\begin{cases} \frac{1}{2} \alpha^T P \alpha + q^T \alpha \rightarrow \min \\ G \alpha \leq h \\ A \alpha = b \end{cases}$$

$$\begin{cases} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j=1}^N y_i y_j \alpha_i \alpha_j x_i^T x_j \\ \alpha_i \geq 0 \\ \sum_{i=1}^N \alpha_i y_i = 0 \end{cases} \rightarrow \begin{array}{ll} P = [y_i y_j x_i^T x_j] & q = [-1] \\ G = -I & h = [0] \\ A = y^T & b = 0 \end{array}$$

Karush-Kuhn-Tucker Conditions

Our (primal) optimization under constraints problem:
$$\begin{cases} \min_z f(z) \\ g_i(z) \leq 0 \\ h_i(z) = 0 \end{cases}$$

If z^* is a local minimum, then exist a Lagrangian multipliers α_i^* and β_j^* for:

$$\mathcal{L}(z, \alpha, \beta) = f(z) + \sum_{i=1}^m \alpha_i g_i(z) + \sum_{j=1}^k \beta_j h_j(z)$$

such that:

$$\begin{cases} \frac{\partial}{\partial z_i} \mathcal{L}(z^*, \alpha^*, \beta^*) = 0 \\ \frac{\partial}{\partial \beta_i} \mathcal{L}(z^*, \alpha^*, \beta^*) = 0 \\ \alpha_i g_i(z^*) = 0 \\ \alpha_i^* \geq 0 \end{cases}$$

And a solution of the primal problem is the solution of the dual problem: $\max_{\alpha, \beta} \mathcal{L}(z, \alpha, \beta)$

Dual Problem

$$\begin{cases} \min_z f(z) \\ g_i(z) \leq 0 \\ h_i(z) = 0 \end{cases} \text{ and } \begin{cases} \frac{1}{2} w^T w \rightarrow \min \\ y_i(w^T x_i - b) \geq 1 \end{cases} \rightarrow \begin{cases} \frac{1}{2} w^T w \rightarrow \min \\ -(y_i(w^T x_i - b) - 1) \leq 0 \end{cases}$$



$$\mathcal{L}(\overbrace{w, b}^z, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i (y_i(w^T x_i - b) - 1)$$

$$\begin{aligned} \alpha_i &\geq 0 \\ \alpha_i (y_i(w^T x_i - b) - 1) &= 0 \end{aligned}$$

Dual Problem: Solution

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i (y_i (w^T x_i - b) - 1) \quad \alpha_i \geq 0, \alpha_i (y_i (w^T x_i - b) - 1) = 0$$

$$\nabla_w \mathcal{L}(w, b, \alpha) = w - \sum_{i=1}^N \alpha_i y_i x_i = 0 \rightarrow w = \sum_{i=1}^N \alpha_i y_i x_i$$

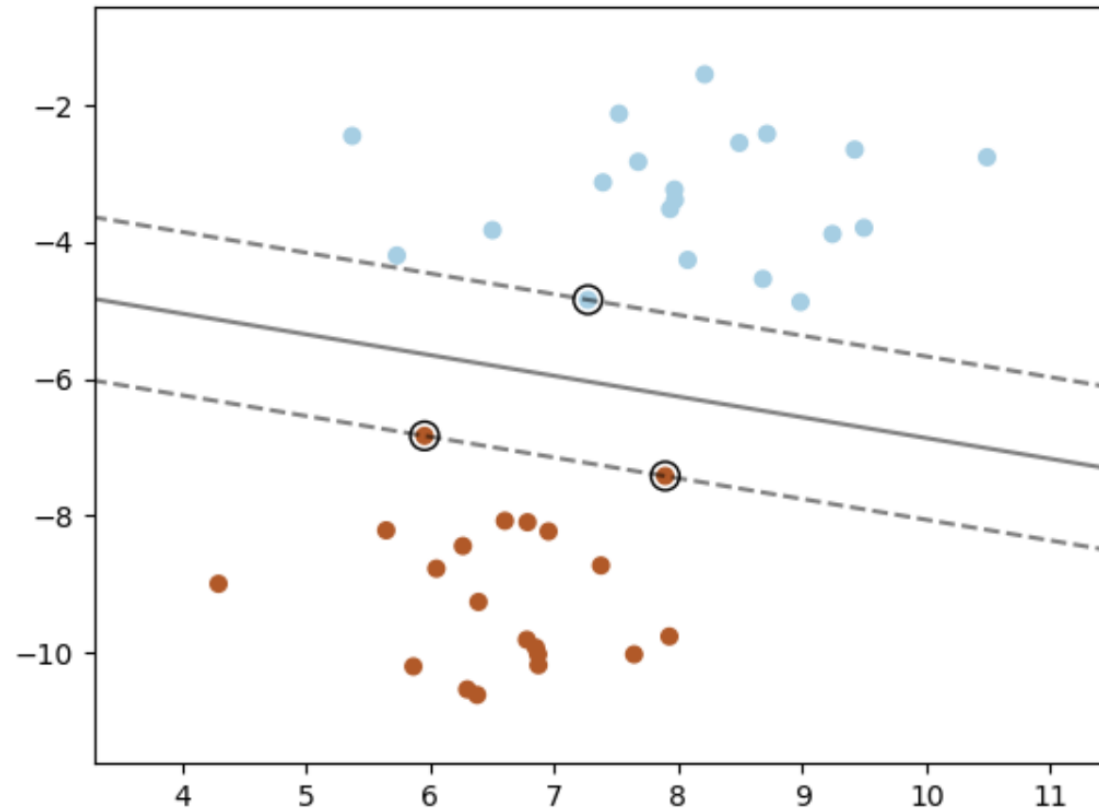
$$\nabla_b \mathcal{L}(w, b, \alpha) = \sum_{i=1}^N \alpha_i y_i = 0$$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \sum_i \sum_j y_i y_j \alpha_i \alpha_j x_i^T x_j - \sum_i \sum_j y_i y_j \alpha_i \alpha_j x_i^T x_j + \sum_i \alpha_i$$

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^N \alpha_i - \sum_{i,j=1}^N y_i y_j \alpha_i \alpha_j x_i^T x_j$$

Quadratic optimization problem under linear constraints is efficiently solved by quadratic programming

Support Vectors



$$w = \sum_{i=1}^N \alpha_i y_i x_i$$

We can find b from $\alpha_i(y_i(w^T x_i - b) - 1) = 0$ where $\alpha_i > 0$

Dual problem solution with CVXOPT Package

Task that QP solver solves:

$$\begin{cases} \frac{1}{2} \alpha^T P \alpha + q^T \alpha \rightarrow \min \\ G \alpha \leq h \\ A \alpha = b \end{cases}$$

$$\begin{cases} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j=1}^N y_i y_j \alpha_i \alpha_j x_i^T x_j \\ \alpha_i \geq 0 \\ \sum_{i=1}^N \alpha_i y_i = 0 \end{cases} \rightarrow \begin{array}{ll} P = [y_i y_j x_i^T x_j] & q = [-1] \\ G = -I & h = [0] \\ A = y^T & b = 0 \end{array}$$

Kernel Trick

Notice, that we only used dot products in all calculations $x_i^T x_j$. That means that we do not need to transition into higher dimensional space but rather only define a dot product operation there.

Definition: we say a function $k(x, x')$ is a **kernel** if it corresponds to a scalar product $\varphi(x)^T \varphi(x')$ in some feature space $\mathbb{R}^M$ (possibly with $M = \infty$; $\phi: X \rightarrow \mathbb{R}^M$)

Mercer's Theorem: a necessary and sufficient condition for a function $k(x, x')$ to be a kernel is that the corresponding Gram matrix $K = \phi\phi^T$ be positive semi-definite for all possible choices of $x_1, \dots, x_n$.

Kernel Trick

Example: let $x \in \mathbb{R}^2$

$$K(x, x') = (1 + x^T x')^2 = \phi^T(x) \phi(x')$$

where

$$\phi(x) = [1 \quad x_1^2 \quad x_2^2 \quad \sqrt{2}x_1 \quad \sqrt{2}x_2 \quad \sqrt{2}x_1x_2]$$

Kernels

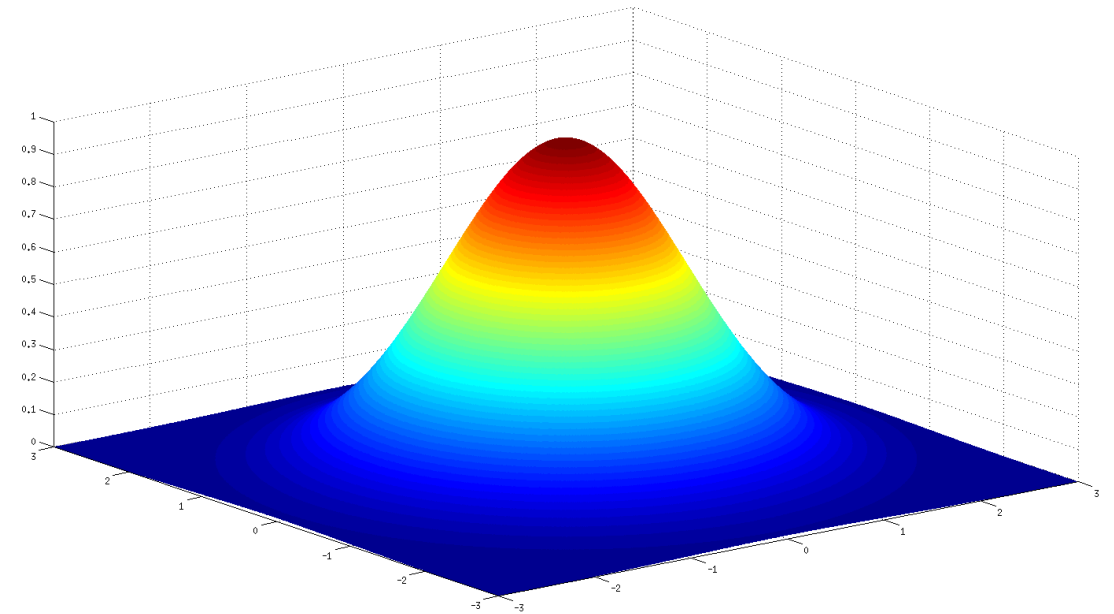
$$\langle x, x' \rangle$$

$$(r + \gamma \langle x, x' \rangle)^d$$

$$e^{-\gamma \|x - x'\|^2}$$

$$\tanh(\gamma \langle x, x' \rangle + r)$$

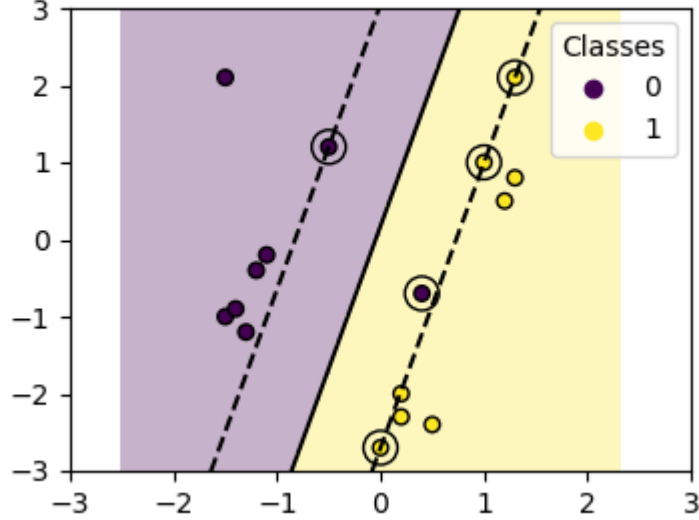
- linear kernel
- polynomial kernel
- radial basis function
- sigmoid kernel



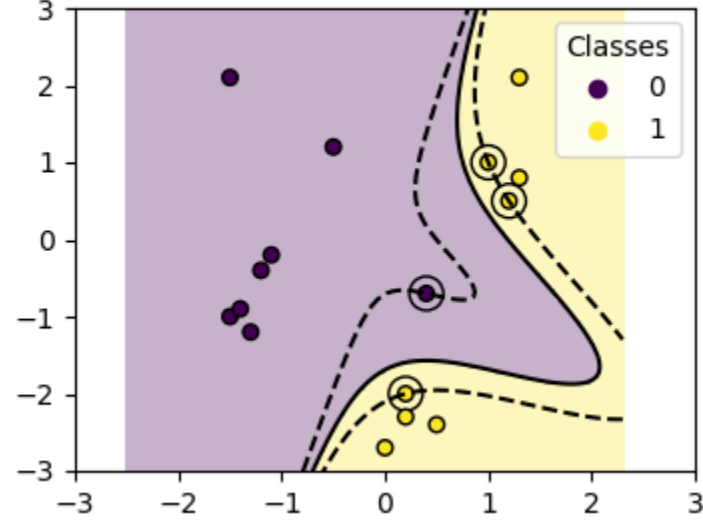
Radial basis function (RBF)

Kernels

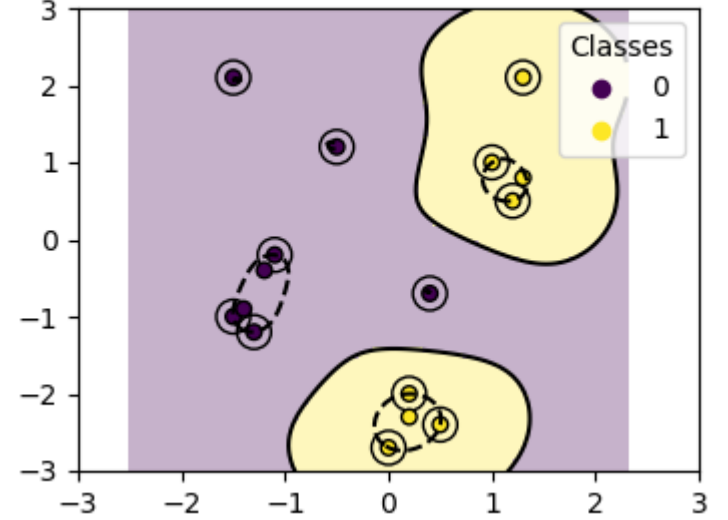
Decision boundaries of linear kernel in SVC



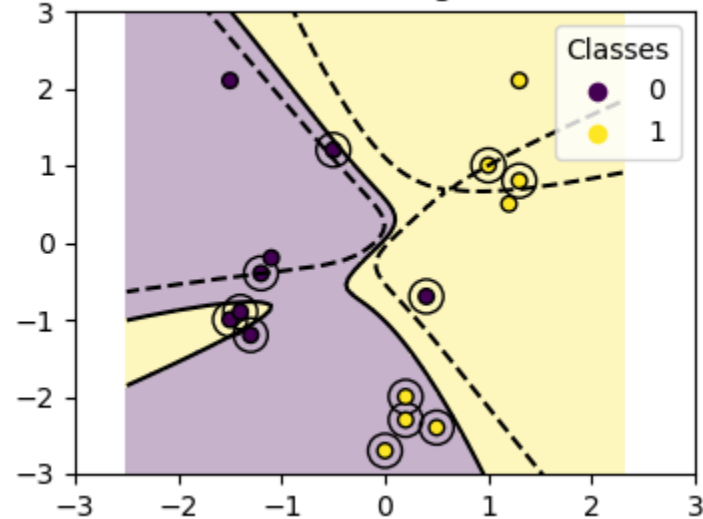
Decision boundaries of poly kernel in SVC



Decision boundaries of rbf kernel in SVC



Decision boundaries of sigmoid kernel in SVC



Soft-Margin SVM

Linearly separable case:

$$\begin{cases} \frac{1}{2}w^T w \rightarrow \min \\ y_i(w^T x_i - b) \geq 1 \end{cases}$$

Linearly inseparable case:

$$\begin{cases} \frac{1}{2}w^T w + C \sum_{i=1}^N \xi_i \rightarrow \min \\ y_i(w^T x_i - b) \geq 1 - \xi_i \\ \xi_i \geq 0 \end{cases}$$

Dual problem for soft margin

$$\begin{aligned}\mathcal{L}(\overbrace{w, b, \xi}^z, \underbrace{\alpha, r}_\alpha) &= \frac{1}{2}w^T w + C \sum_{i=1}^N \xi_i - \sum_{i=1}^N \alpha_i (y_i (w^T x_i - b) - 1 + \xi_i) - \sum_{i=1}^N r_i \xi_i \\ &= \frac{1}{2}w^T w - \sum_{i=1}^N \alpha_i (y_i (w^T x_i - b) - 1) - \sum_{i=1}^N \xi_i (r_i + \alpha_i - C) \\ \alpha_i, r_i &\geq 0; \quad \alpha_i (y_i (w^T x_i - b) - 1 + \xi_i) = 0; \quad r_i \xi_i = 0\end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial b} = \frac{\partial \mathcal{L}}{\partial \xi} = 0 \Rightarrow \begin{cases} w = \sum \alpha_i y_i x_i \\ \sum \alpha_i y_i = 0 \\ \alpha_i = C - r_i \Rightarrow 0 \leq \alpha_i \leq C \end{cases}$$

Dual problem for soft margin

$$\left\{ \begin{array}{l} w = \sum \alpha_i y_i x_i \\ \sum \alpha_i y_i = 0 \\ \alpha_i = C - r_i \Rightarrow 0 \leq \alpha_i \leq C \end{array} \right. \Rightarrow$$

$$\begin{aligned} \Rightarrow \mathcal{L}(\alpha) &= \frac{1}{2} w^T w - \sum \alpha_i (y_i (w^T x_i - b) - 1) - \sum \xi_i (r_i + \alpha_i - C) = \\ &= \frac{1}{2} \sum \sum y_i y_j \alpha_i \alpha_j x_i^T x_j - \sum \sum y_i y_j \alpha_i \alpha_j x_i^T x_j + \sum \alpha_i \end{aligned}$$

Dual problem for soft margin

$$\left\{ \begin{array}{l} \max_{\alpha} \mathcal{L}(\alpha) = \sum \alpha_i - \frac{1}{2} \sum \sum y_i y_j a_i \alpha_j x_i^T x_j \\ \sum \alpha_i y_i = 0 \\ 0 \leq \alpha_i \leq C \end{array} \right.$$

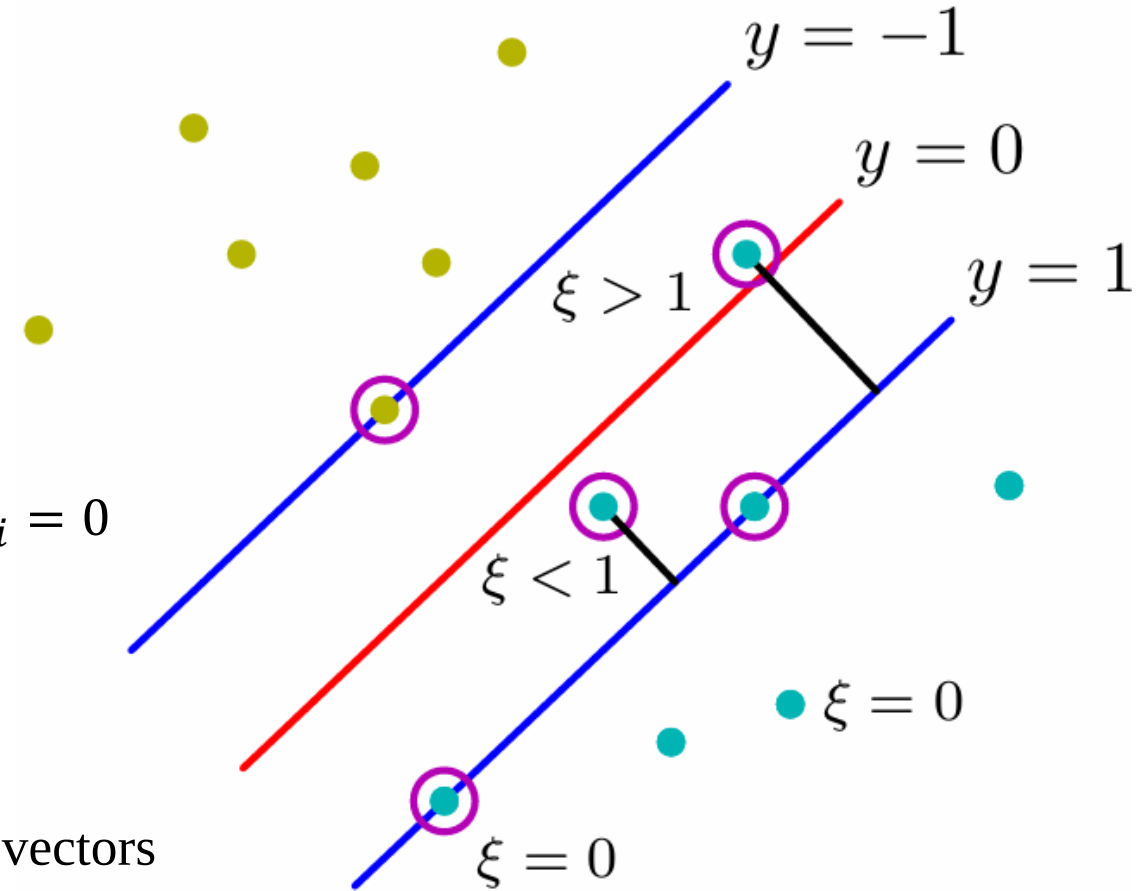
Vector Types

$$w = \sum \alpha_i y_i x_i$$

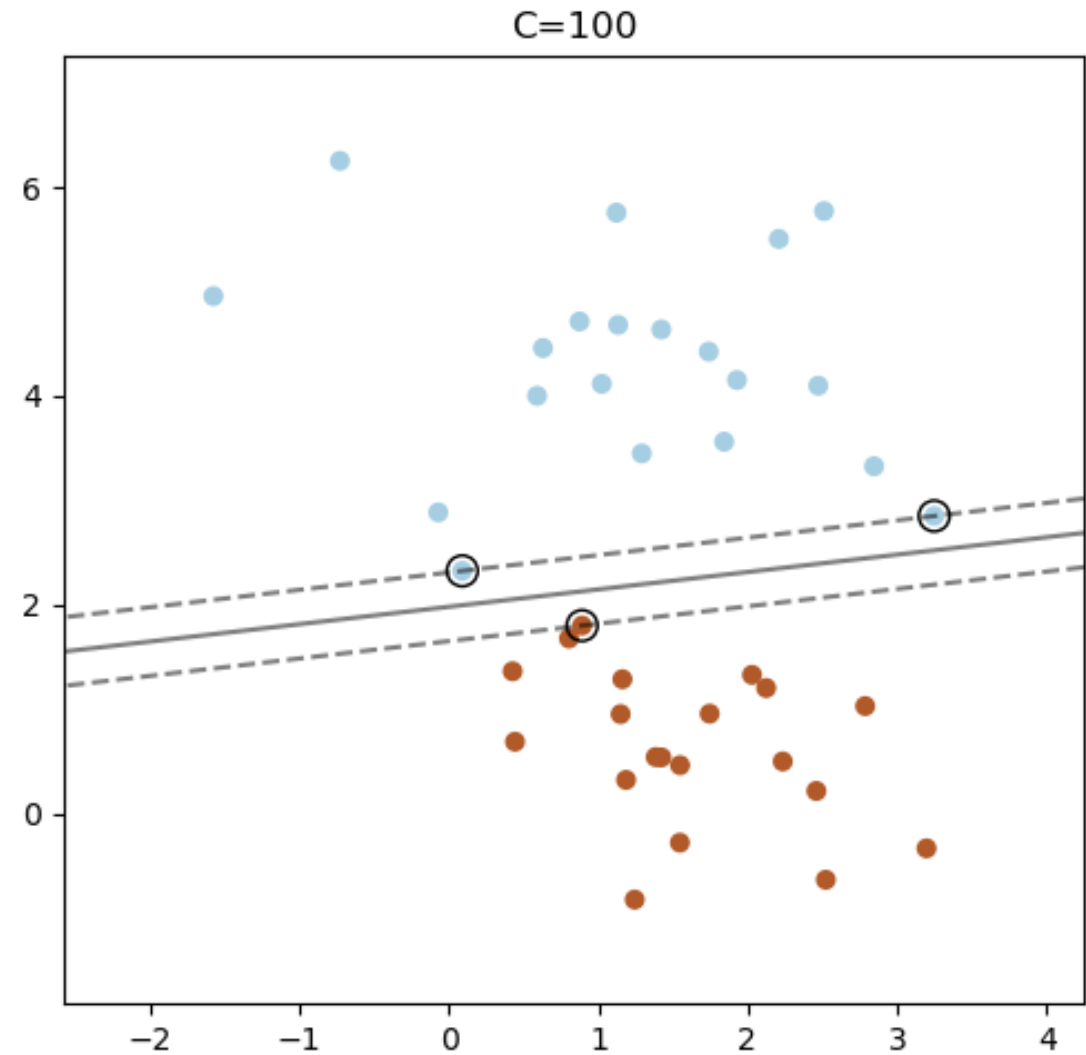
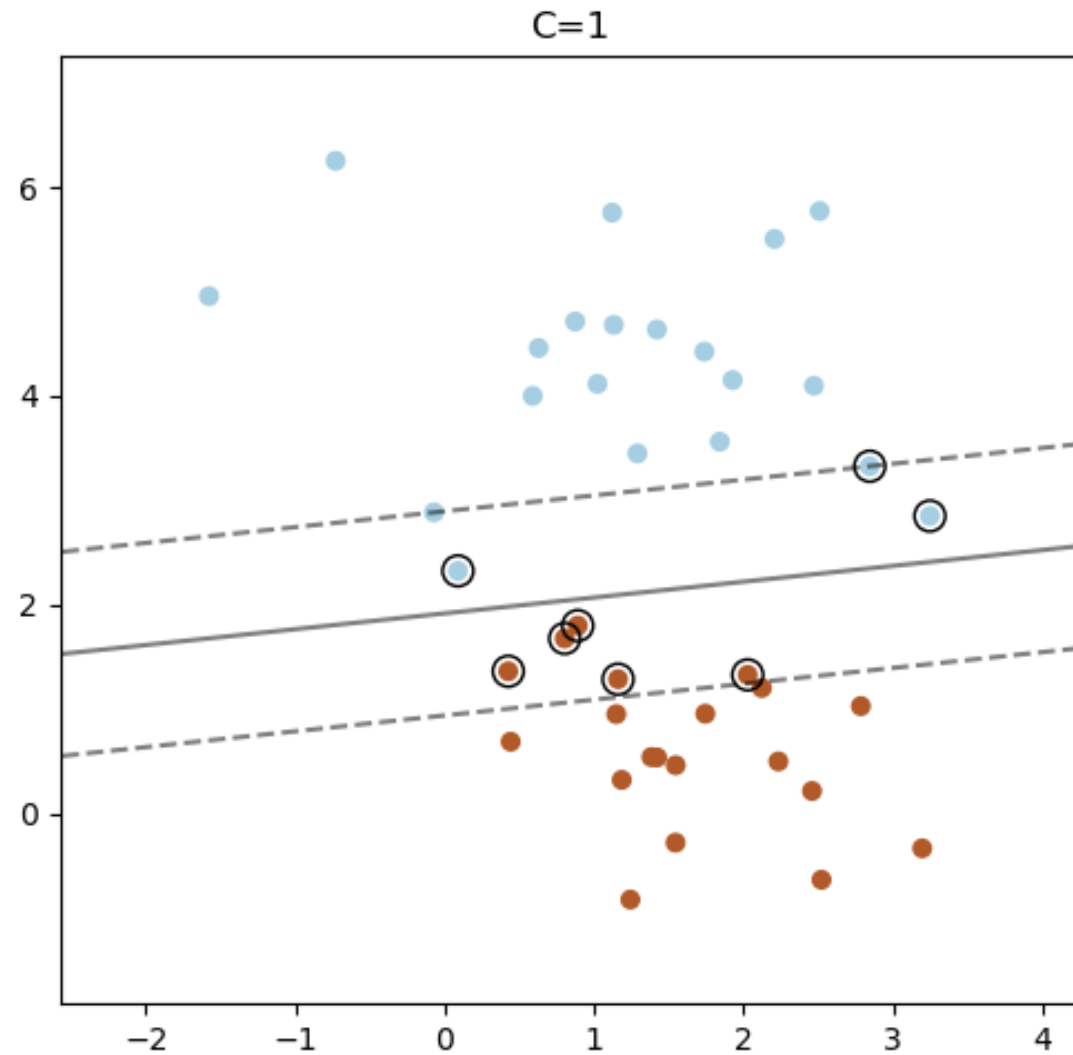
$$y_i(w^T x_i - b) \geq 1 - \xi_i$$

$$\alpha_i = C - r_i; \quad \alpha_i (y_i(w^T x_i - b) - 1 + \xi_i) = 0; \quad r_i \xi_i = 0$$

1. $\alpha_i = 0, \xi_i = 0, y_i(w^T x_i - b) \geq 1$ – Inside vectors
2. $0 < \alpha_i < C, \xi_i = 0, y_i(w^T x_i - b) = 1$ – “Good” support vectors
3. $\alpha_i = C, \xi_i > 0, y_i(w^T x_i - b) \leq 1$ – “Bad” support vectors

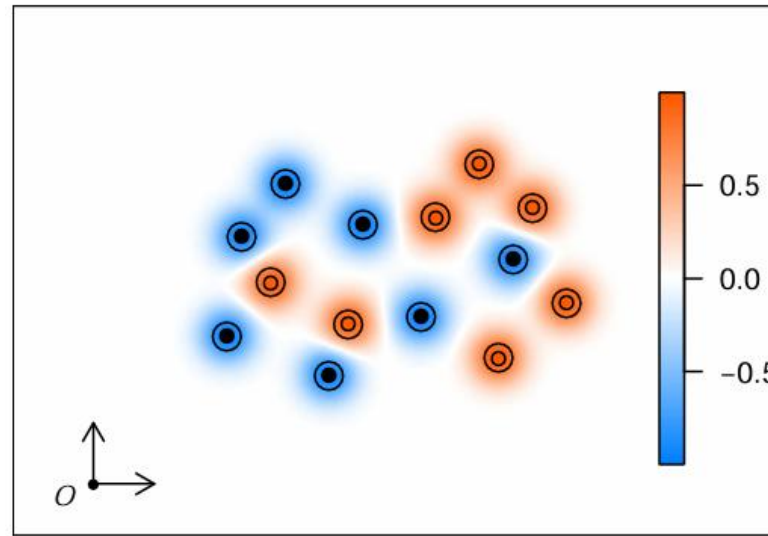


Soft-Margin Coefficient

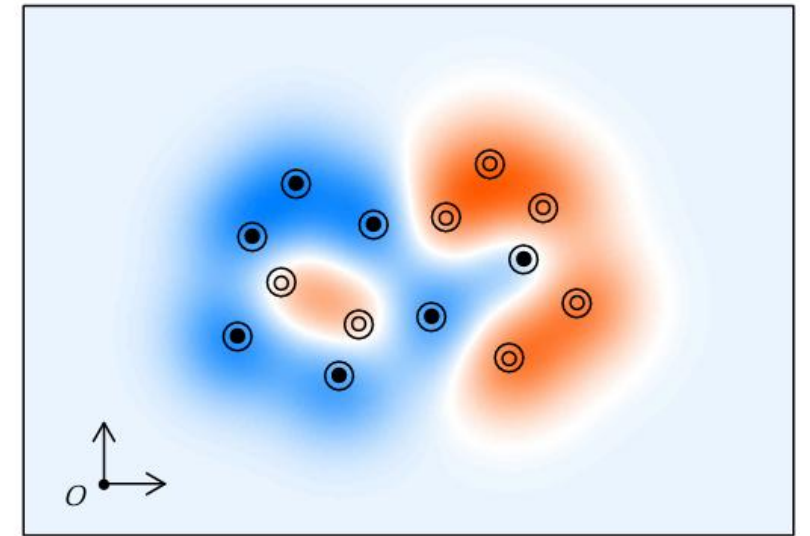


Higher dimensionality and generalization

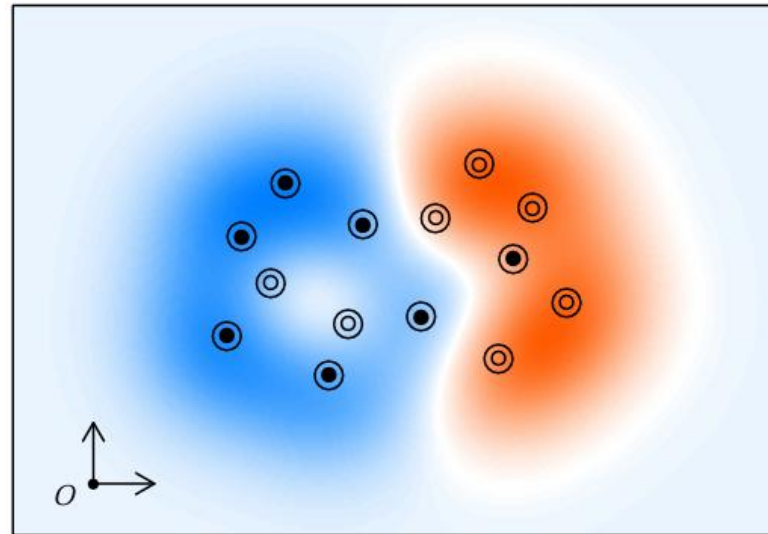
$$E_{gen} \leq \frac{\# \text{ of support vectors}}{\# \text{ of training vectors}}$$



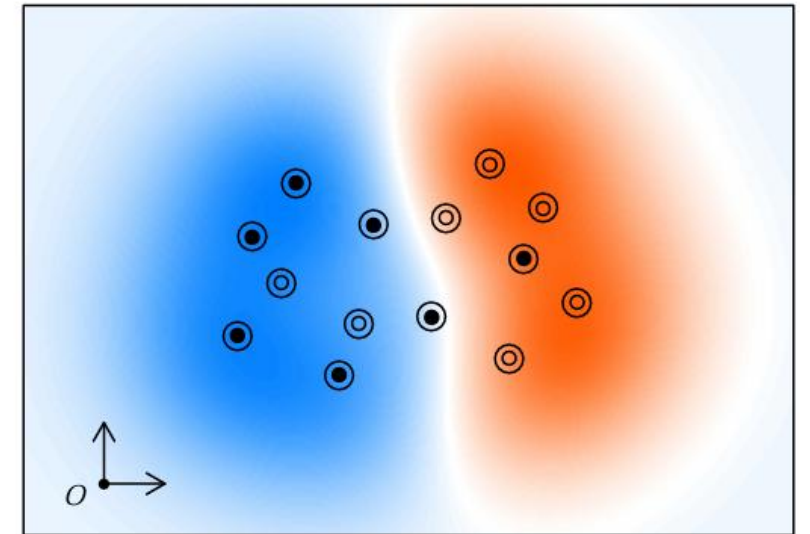
$\sigma = 3.2$



$\sigma = 7.1$



$\sigma = 10.0$

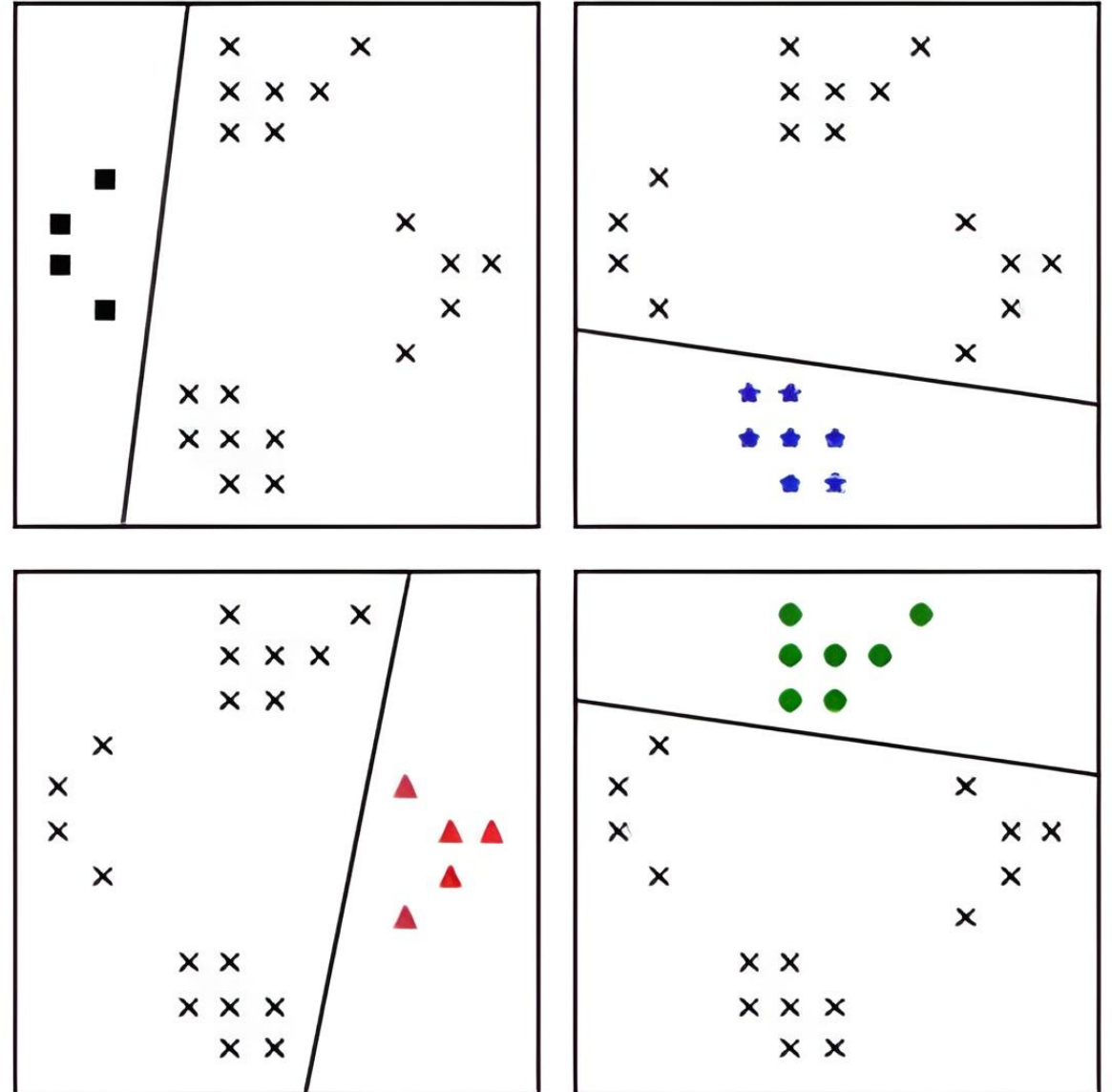


$\sigma = 14.1$

Multi-Class SVM

One-vs-all: train K different SVM's with the k^{th} SVM trained on C_k versus *not* C_k . Then $y(x) = \max_k y_k(x)$.

One-versus-one: train all $\frac{K(K-1)}{2}$ two-class SVMs and then use majority voting to obtain final classifier



Support Vector Regression Machine

$$\left\{ \begin{array}{l} \frac{1}{2} \|w\|^2 + C \sum (\xi_i + \xi_i^*) \rightarrow \min \\ y_i - w^T x_i - b \leq \epsilon + \xi_i \\ w^T x_i + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{array} \right.$$

